

ORBIT — OFFLINE MULTIMODAL RAG ASSISTANT

COMPLETE USER MANUAL, INSTALLATION & FEATURE GUIDE

1. Executive Overview & Mission

Orbit is an air-gapped, local-first Retrieval-Augmented Generation (RAG) platform. It provides instant semantic Q&A, voice conversations, document quizzes, mind maps, and report generation over private documents without sending any data to cloud services.

Platform Name	Orbit (Offline Multimodal RAG Assistant)
Live Website Portal	https://offlinerag-psi.vercel.app
Direct Download Link	https://github.com/kalakonda-akshay/RAG/releases/latest/download/OfflineRAGAssistant_Setup.zip
Supported Formats	29 Extensions (PDF, Word, Excel, PPTX, Audio, Video, Images, SQL)

2. Download & Installation Step-by-Step Guide

Step 1: Download Setup Package

Download *OfflineRAGAssistant_Setup.zip* (1.9 GB) from <https://offlinerag-psi.vercel.app> or directly from GitHub Releases.

Step 2: Extract & Environment Setup

Extract the zip archive and run the setup script: `python launcher/first_run_setup.py`

Step 3: Pull Local LLM & Embedding Models

Run in terminal: `ollama pull llama3.2:3b` and `ollama pull nomic-embed-text`

Step 4: Launch Web Interface

Start Orbit: `python launcher/run_app.py` → Open browser at **`http://localhost:8501`**

3. Visual Screenshot Tour

Figure 1: Grounded Q&A; Thread & Interactive Citations

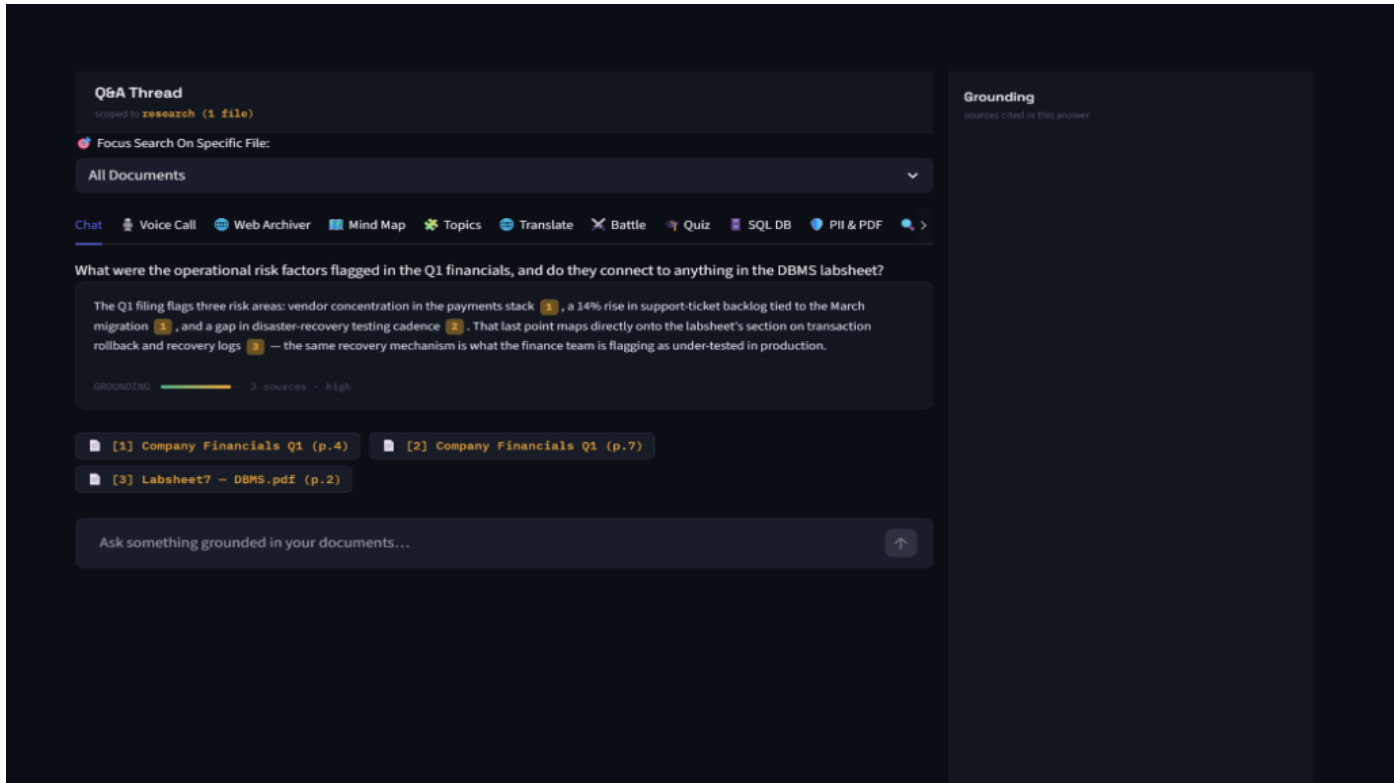


Figure 2: Real-Time Voice Call Mode

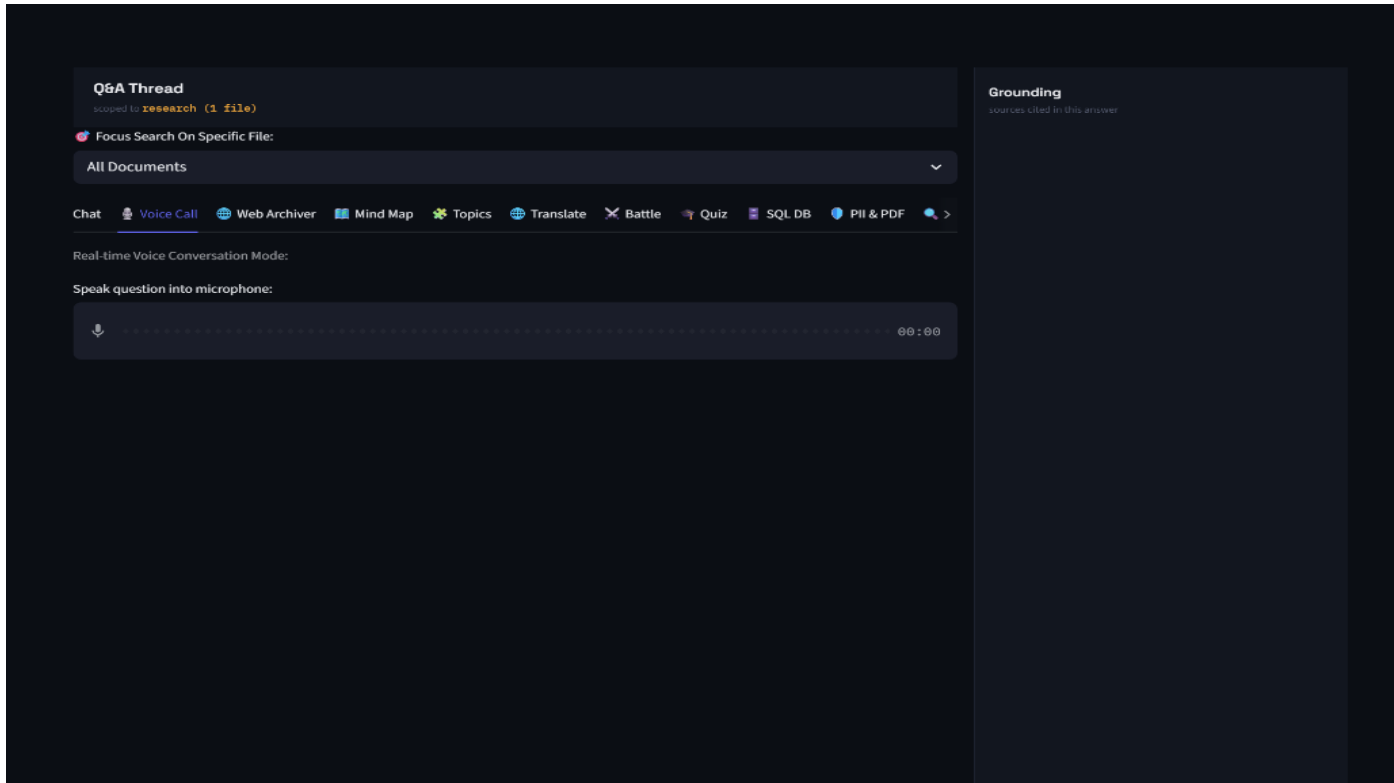
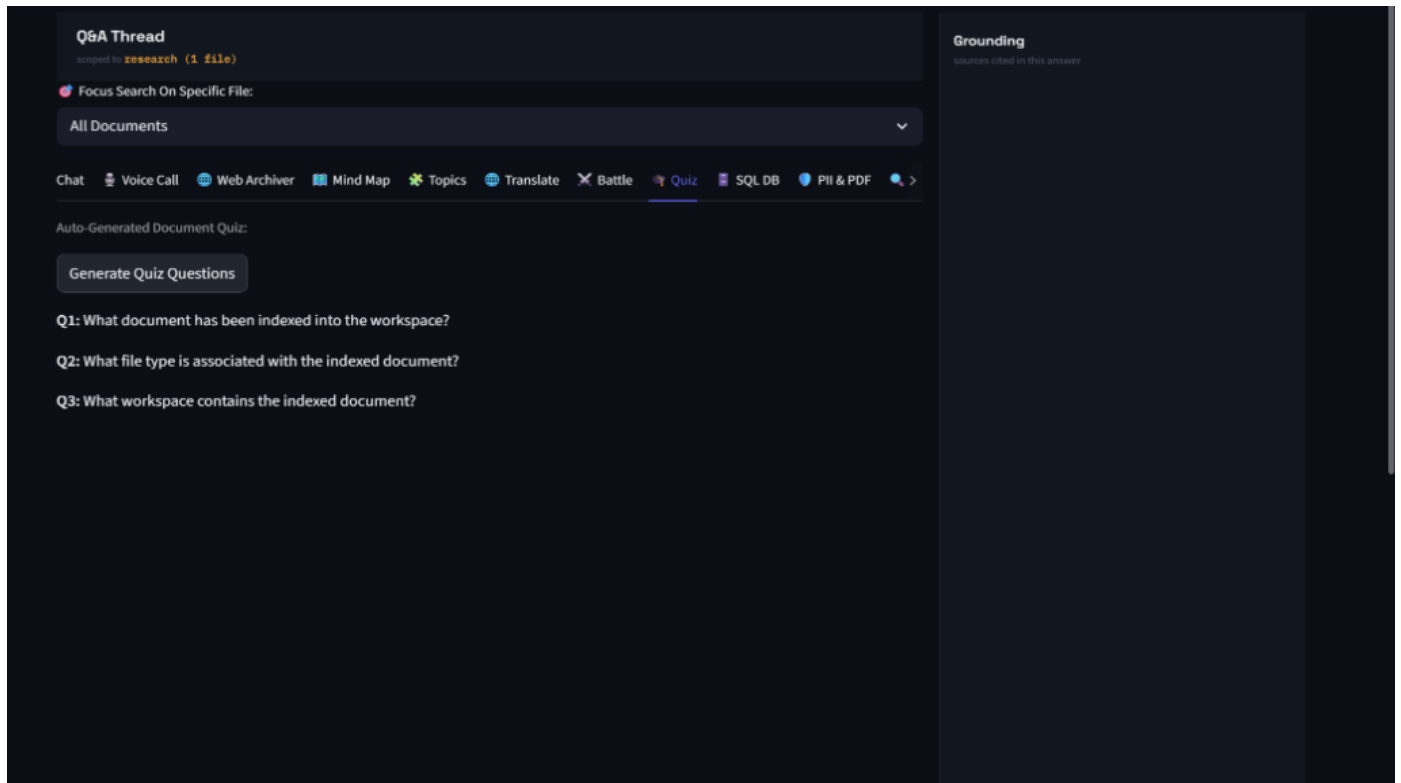


Figure 3: Auto-Generated Document Quiz Workspace



4. Complete Feature Reference (12 Tabs)

- **Grounded Q&A; Chat:** Dense vector search + Rank-BM25 hybrid retrieval + Self-RAG verification.
- **Voice Call Mode:** Real-time microphone recording and Whisper speech transcription.
- **Web Archiver:** Parses and indexes offline web pages and documentation HTML.
- **Mind Map Generator:** Generates visual Mermaid.js mind maps from document concepts.
- **Topic Cluster Discovery:** Performs K-Means semantic clustering across document text.
- **Multilingual Translator:** Translates document passages into 10+ target languages offline.
- **Model Battle Mode:** Compares responses side-by-side across multiple local LLMs.
- **Auto Document Quiz:** Generates practice test questions with answer keys.
- **Natural Language SQL Engine:** Queries SQLite (.db/.sqlite) tables using auto-generated SQL.
- **PII Redactor & PDF Exporter:** Redacts SSNs, credit cards, emails, and exports sanitized PDFs.
- **Document Comparator:** Side-by-side comparison of two documents highlighting diffs.
- **Report & PPTX Exporter:** 1-click PowerPoint presentations (.pptx) and executive brief summaries.